IMO Phase II Level 2
Probabilistic Method

Omega Tong

2nd October, 2021

1 Linearity of Expectation

Definition 1.1 (Random Variable)

A **random variable** is a variable whose value is unknown or a function that assigns values to each of an experiment's outcomes.

Definition 1.2 (Expected Value)

The **expected value** is the "weighted average" of a random variable X:

$$\mathbb{E}[X] := \sum_x \mathbb{P}(X = x) \cdot x.$$

Example 1.3

There are n people, each of who has a name tag. We shuffle the name tags and randomly give each person one of the name tags. Let S be the number of people who receive their own name tag. Prove that the expected value of S is 1. (i.e. S independent of n)

Proof. We can list all the $n!$ permutations and count the number of cases where the i-th person get his own name tag. We can easily see that there are $(n-1)!$ cases where the i-th person get his own tag. Hence we have $\mathbb{E}[S] = \frac{1}{n!}\left((n-1)! * n\right) = 1$. □

Theorem 1.4 (Linearity of Expectation)

Given any random variables $X_1, X_2, \ldots, X_n$, we always have

$$\mathbb{E}[X_1 + X_2 + \cdots + X_n] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \cdots + \mathbb{E}[X_n].$$

Remark. It is true even if X_is are dependent of each other.

Proof.

$$\begin{aligned}
\mathbb{E}[X+Y] &= \sum_x \sum_y [(x+y) \cdot P(X = x, Y = y)] \\
&= \sum_x x \sum_y P(X = x, Y = y) + \sum_y y \sum_x P(X = x, Y = y) \\
&= \sum_x x \cdot P(X = x) + \sum_y y \cdot P(y = y) \\
&= \mathbb{E}[X] + \mathbb{E}[Y].
\end{aligned}$$

□

Example 1.5

At a nursery, 2006 babies sit in a circle. Suddenly, each baby randomly pokes either the baby to its left or to its right. What is the expected value of the number of unpoked babies?

Proof. Number the babies $1, 2, \ldots, 2006$. Define

$$X_i := \begin{cases} 1, & \text{if baby } i \text{ is unpoked} \\ 0, & \text{otherwise.} \end{cases}$$

We seek $\mathbb{E}[X_1 + X_2 + \cdots + X_{2006}]$. Note that any particular baby has probably $(\frac{1}{2})^2 = \frac{1}{4}$ of being unpoked. Hence $\mathbb{E}[X_i] = \frac{1}{4}$ for each i and hence

$$\mathbb{E}[X_1 + \cdots + X_2 + \cdots + X_{2006}] = \sum_{i=1}^{2006} \mathbb{E}[X_i] = 2006 \cdot \frac{1}{4} = \frac{1003}{2}.$$

□

Usage 1.6 (Use Expected Value to Show Existence)

Suppose we know the average score of a test is 12.1, then there exists a student who got at least 13 points, and a contestant who got at most 12 points.

Remark. It is similar in spirit to the pigeonhole principle.

Example 1.7

Prove that any subgraph of $K_{n,n}$ with at least $n^2 - n + 1$ edges has a perfect matching.

Proof. We randomly pair off one set of points with the other (whether there is an actual edge or not), and define the score of such a pairing be the number of pairs which are actually connected by ane edge. Number the pairs by $1, 2, \ldots, n$. Define

$$X_i := \begin{cases} 1 & \text{if the } i\text{th pair is connected by an edge} \\ 0 & \text{otherwise.} \end{cases}$$

Hence the score of a config $X = X_1 + \cdots + X_n$. For any pair of points, they are connected with probability at least $\frac{n^2-n+1}{n^2}$. Hence $\mathbb{E}[X_i] = \frac{n^2-n+1}{n^2}$, and hence

$$\begin{aligned} \mathbb{E}[X] &= \mathbb{E}[X_1] + \cdots + \mathbb{E}[X_n] = n \cdot \mathbb{E}[X_1] \\ &= \frac{n^2 - n + 1}{n} \\ &= n - 1 + \frac{1}{n}. \end{aligned}$$

Since X takes only integer values, there exists a config with $X = n$. □

Theorem 1.8 (Jenson's Inequality)

If f is a convex function and $t \in [0, 1]$, we have

$$f(tx_1 + (1-t)x_2) \leq tf(x_1) + (1-t)f(x_2).$$

In context of probability theory, if X is a random variable and φ is a convex function, then

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)].$$

2 Problems for Discussion

Problem 1. (IMC 2002 day2 q2) 200 students participated in a math contest. They had 6 problems to solve. Each problem was correctly solved by at least 120 participants. Prove that there must be 2 participants such that every problem was solved by at least one of these two students.

Problem 2. (Russia 1996) In the Duma there are 1600 delegates, who have formed 16000 committees of 80 persons each. Prove that one can find two committees having no fewer than four common members.

Problem 3. (BAMO 2004) Consider n real numbers, not all zero, with sum zero. Prove that one can label the numbers as $a_1, a_2, \ldots, a_n$ such that $a_1a_2 + a_2a_3 + \cdots + a_na_1 < 0$.

Problem 4. (IMOSL 1999 C4) Let A be any set of n residues mod n^2. Show that there is a set B of n residues mod n^2 that at least half of the residues mod n^2 can be written as $a + b$ with a in A and b in B.

Problem 5. (Online Math Open q29) Kevin has $2^n - 1$ cookies, each labeled with a unique nonempty subset of $\{1, 2, \ldots, n\}$. Each day, he chooses one cookie uniformly at random out of the cookies not yet eaten. Then, he eats that cookie, and all remaining cookies that are labeled with a subset of that cookie. Compute the expected value of the number of days that Kevin eats a cookie before all cookies are gone.

Problem 6. (EGMO 2019 P5) Let $n \geq 2$ be an integer, and let $a_1, a_2, \cdots, a_n$ be positive integers. Show that there exist positive integers $b_1, b_2, \cdots, b_n$ satisfying the following three conditions:

(A) $a_i \leq b_i$ for $i = 1, 2, \cdots, n$;

(B) the remainders of $b_1, b_2, \cdots, b_n$ on division by n are pairwise different; and

(C) $b_1 + b_2 + \cdots b_n \leq n\left(\frac{n-1}{2} + \left\lfloor \frac{a_1+a_2+\cdots a_n}{n} \right\rfloor\right)$

Problem 7. (IMC 2017 q4) There are n people in a city, and each of them has exactly 1000 friends (friendship is always symmetric). Prove that it is possible to select a group S of people such that at least $\frac{n}{2017}$ persons in S have exactly two friends in S.

Problem 8. (Iran TST 2008 P6) Prove that in a tournament with 799 teams, there exist 14 teams, that can be partitioned into groups in a way that all of the teams in the first group have won all of the teams in the second group.

Problem 9. (Caro-Wei Theorem) Consider a graph G with vertex set V . Prove that one can find an independent set of V (pairwise non-adjacent) with size at least

$$\sum_{v \in V} \frac{1}{\deg v + 1}.$$

Remark. By applying Jensen's inequality, our independent set has size at least $\frac{n}{d+1}$, where d is the average degree. This result is called Turan's Theorem.